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The calculated parameters needed for the circuit design are described in Fig.1 and Fig.2

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The given data's are:-

$$\begin{aligned} \ell &= 3 \cdot X \\ t_d &= 1 \text{ ns} \cdot X \\ R_n &= 4.5 \end{aligned}$$

$$\begin{aligned} X &= 1 + n/5 \quad | n=0 \\ \boxed{X=1} \end{aligned}$$

So, Length of the line is : $3 \cdot X = 3 \text{ inches} = 0.0762 \text{ m}$

Rise time of the signal is : $(1 \times 1) \text{ ns} = 1 \text{ ns}$

Also, the impedance of the transmission line, $Z_0 = 60 \Omega$.

So, the wave velocity is $\Rightarrow V = \frac{c}{\sqrt{\epsilon_r}}$ (This is also known as propagation speed J).
 $= 141.42 \times 10^6 \text{ m/s}$

The total inductance L and capacitance C of the transmission line can be calculated from the characteristic impedance Z_0 and propagation delay t_d :-

$$Z_0 = \sqrt{\frac{L}{C}} \quad \& \quad t_d = \sqrt{LC} \quad \left| \begin{array}{l} \text{Here, } \frac{\ell}{t_d} = \frac{L}{C} \\ = \frac{0.0762 \text{ m}}{141.42 \times 10^6 \text{ m/s}} \\ = 5.38 \times 10^{-10} \text{ sec} \\ = 538.8 \text{ psec} \end{array} \right.$$

$$\begin{aligned} \therefore L &= Z_0 \cdot t_d = 60 \times 538.8 \text{ psec} \\ &= 32.32 \text{ nH} \end{aligned}$$

$$C = \frac{t_d}{Z_0} = \frac{538.8 \text{ psec}}{60} = 8.98 \text{ pF}$$

Fig.1: Calculation for the circuit diagram

The minimum number of segment is:-

$$n_{min} > 10 \cdot \frac{1}{\Delta t \cdot v} \quad \text{where } \Delta t = 1 \mu s$$

$$n_{min} > 10 \cdot \frac{t_d}{\Delta t} = 10 \times \frac{538.8 \text{ psec}}{1 \text{ ns}}$$

$$= 5.38$$

$$\approx 6$$

Now, for each each circuit segment inductance $2L$, capacitance C_{SEG} are,

$$L_{SEG} = \frac{L}{n_{min}} = \frac{32.32 \text{ nH}}{6} = 5.38 \text{ nH}$$

$$C_{SEG} = \frac{C}{n_{min}} = \frac{8.98 \text{ pF}}{6} = 1.49 \text{ pF}$$

Fig. 2: Calculation for the circuit diagram

Now the circuit design is given below:

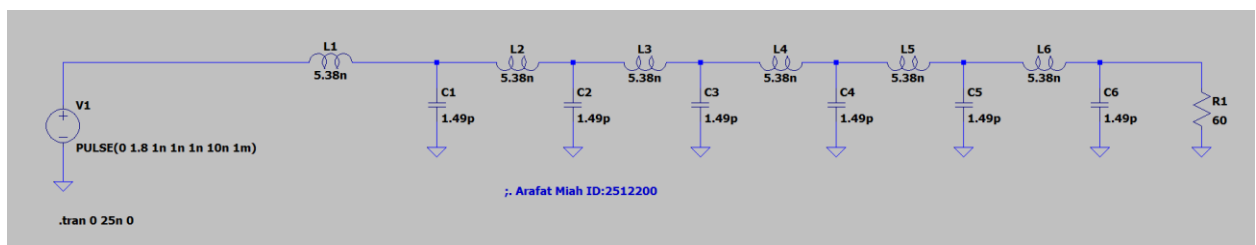


Fig 3: Design of the lossless RLGC circuit.

The input and output curve is shown below:

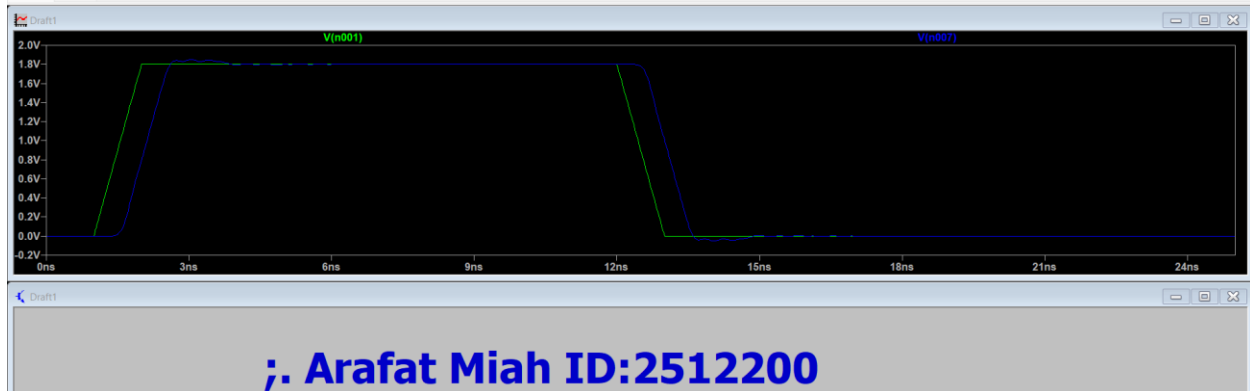


Fig. 4: Input and output curve of the lossless RLGC circuit.

Now the measured propagation delay of the circuit is given below:



Fig. 5: Propagation delay between input and output.

Observation: We can clearly see that the measured propagation delay of the circuit is |561.33625ps| which is almost same as I calculated $T_d = 538.8\text{psec}$

So the results matched, which indicates the design works correctly.

And if we calculate the registrant value from the input side of the design from LTspice directly, it will prove that it is a lossless transmission line. Fig. 6 provides the resistance value directly from the simulation.

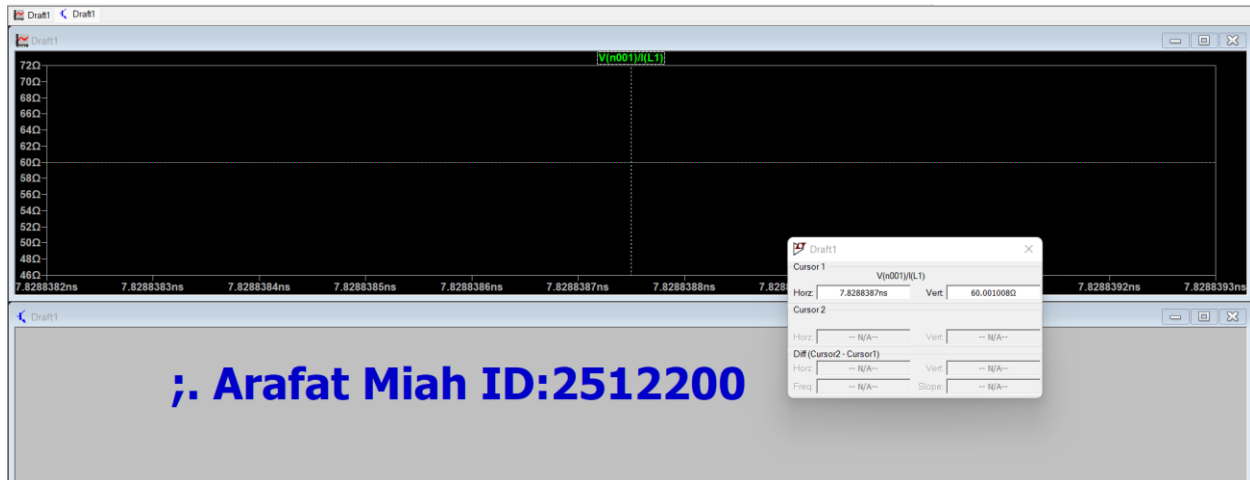


Fig. 6: Resistance at input side.

Observation: Here we can see that the resistance value is 60Ω which is exactly same as the R_1 , so the system is lossless.

Now we will compare the designed lossless RLGC circuit with the ideal transmission line.

Here is the ideal transmission line with the $T_d = 538.8\text{ps}$ and $Z_0 = 60\Omega$ and the $R_{\text{load}} = 60\Omega$.

We can see that I have chosen the $Z_0 = R_{\text{load}}$, that is because that will mitigate the reflection voltage.

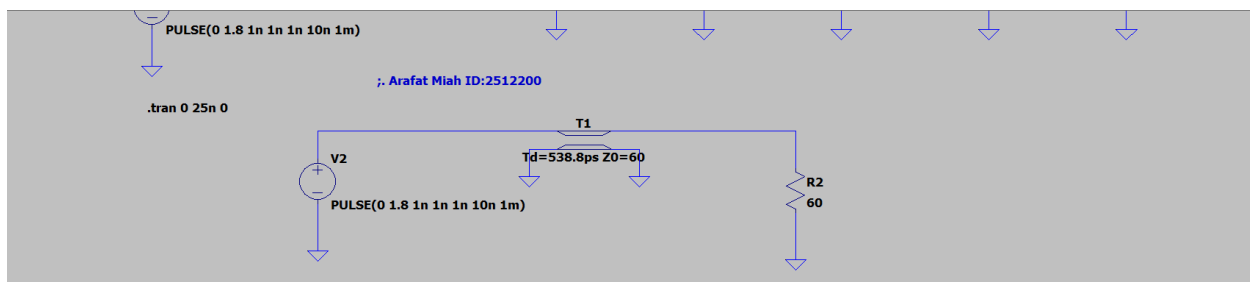


Fig. 7: Ideal transmission line.

The simulation result of this circuit is :

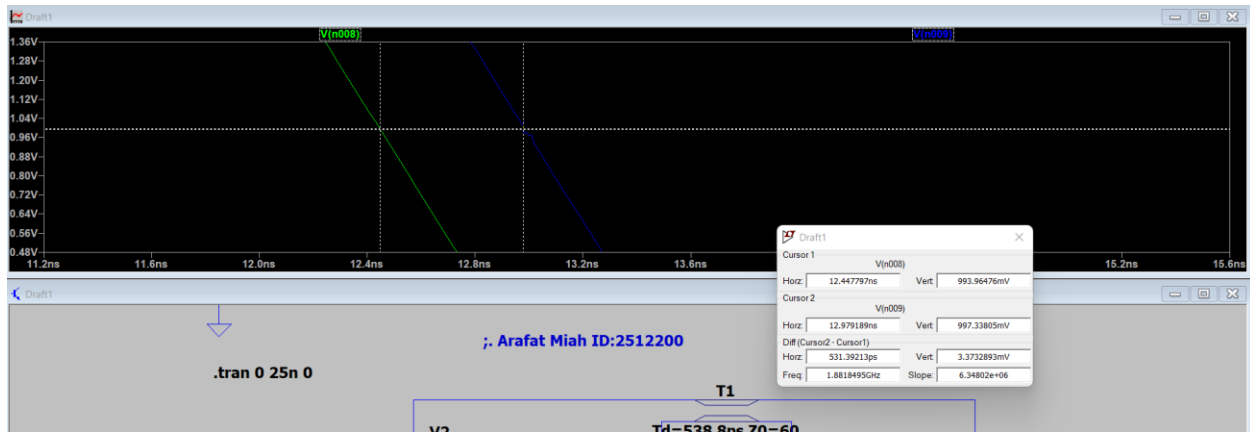


Fig. 8: Input and output voltage curve of the ideal transmission line with showing Time Delay.

Observation: We can clearly see that the simulation value T_d calculated from my design circuit and the time delay from the ideal transmission line almost matched perfectly. (see Fig. 8 and Fig. 5)

Last question from the assignment: what effect the number of RLCG-elements have?

Ans: In this design we neglect the effect of series resistance and shunt resistance to make the circuit lossless, so the signal entering at one end of the line will exit the other end, with no change in amplitude or shape. That is proven in the above simulations.